Notes: Choi, Tian, Wu, & Kargar (JFE, 2025)

Investor demand, firm investment, and capital misallocation

Contents

1 Big picture

- **Question.** Do fluctuations in investor demand for a firm's stock affect real investment, and do they distort the allocation of capital across firms?
- **Core contribution.** The paper embeds a demand system for equity investors into a dynamic corporate finance model in which firms choose investment, debt, cash, dividends, and equity issuance or repurchase. The key novelty is that investor demand is *endogenous to firm characteristics*, and firm policies feed back into those same characteristics.
- **Economic mechanism.** Investor demand generates two opposing forces. First, high demand relaxes financing constraints and can push firms closer to their unconstrained investment level (*constraint-easing effect*). Second, heterogeneous demand across firms creates dispersion in financing conditions and induces firms to time the market, which can crowd out investment and widen dispersion in marginal products of capital (*dispersion effect*).
- **Data.** The paper combines CRSP-Compustat firm data with granular institutional holdings from Form 13F. The sample runs from 1980 to 2021 for the reduced-form evidence and uses 173,238 firm-year observations with institutional holdings in the structural estimation.
- **Identification strategy.** Demand-side parameters are estimated by *indirect inference*: the model is simulated until the endogenous relationship between investor portfolio weights and firm characteristics matches the auxiliary regression in the data. Firm-side parameters are estimated by simulated method of moments.
- **Main firm-level result.** Strong demand raises stock prices, encourages equity issuance, increases cash, and boosts investment. Weak demand encourages opportunistic repurchases and especially crowds out investment for financially constrained firms.
- **Main aggregate result.** Turning off excess investor demand lowers the dispersion of log(MPK) by 26.9% and reduces TFP losses by 23.36%. Thus, the dispersion effect dominates the average constraint-easing effect in the aggregate.
- **Additional insight.** Shifts in investor clientele that resemble the 2021 "meme stock" episode dramatically increase valuation and issuance, but the marginal return on capital falls, implying overinvestment in subsidized firms.
- **Interpretation of excess demand.** The paper links model-implied latent demand more strongly to *non-fundamental* asset-pricing factors and direct sentiment proxies than to fundamental accounting factors, supporting a sentiment-based interpretation.
- **Why the paper matters.** The paper shows that investor demand is not merely an asset-pricing object. It changes the real allocation of physical capital and is quantitatively comparable to major misallocation channels already emphasized in macro-finance.

2 Data and measurement (key objects and motivating facts)

2.1 Corporate and holdings data

- **Corporate data.** Stock prices come from CRSP and accounting data come from Compustat.
- **Investor holdings.** Institutional holdings come from the Thompson Reuters Institutional Holdings Database based on Form 13F filings.

- **Reporting threshold.** U.S. institutions with assets under management above \$100 million must disclose long positions when a holding exceeds either 10,000 shares or \$200,000.
- **Merged estimation sample.** The structural estimation sample contains 173,238 firm-year observations and 17,130 distinct firms after merging holdings with CRSP-Compustat and dropping unmatched observations.
- **Sample restrictions.** The main empirical sample excludes utilities (SIC 4900–4999) and financial firms (SIC 6000–6999).
- **Portfolio weights.** For institution i, the portfolio weight on stock n is the dollar amount invested in that stock divided by the total dollar amount invested in all equities held by the institution.
- **Outside asset.** Stocks with missing characteristics or returns, or with CRSP share codes other than 10 or 11, are classified as the outside asset, following Koijen and Yogo (2019).
- **Investment universe.** The investment universe consists of stocks held in the current quarter and previous 12 quarters.

2.2 Main variables from Table 1

- **Real-side outcomes.** Gross investment is capital expenditures minus sale of property, scaled by gross PPE; income-to-assets is operating income before depreciation divided by assets.
- **Financing variables.** Equity issuance is sales of common and preferred stock scaled by assets when issuance exceeds 2%; debt issuance is the change in total debt scaled by assets when positive debt issuance exceeds 2%; share repurchase is purchases of common and preferred stock scaled by assets.
- **Valuation and demand shifters.** Market-to-book equals total assets minus book equity plus market equity, divided by total assets; dividend yield is dividends divided by market equity; price changes are annual stock returns.
- **Mispricing proxies.** DACCR is discretionary accruals following Polk and Sapienza (2008) and Chan et al. (2001).
- **Equity dependence proxies.** KZ is the four-variable Kaplan-Zingales index, FF is the financial flexibility index of Doidge et al. (2009), and ED is the rolling five-year average equity issuance minus debt issuance ratio.

2.3 Summary statistics (Table 3)

- **Typical firm.** Mean income-to-assets is 0.1367, mean leverage is 0.2367, mean firm size is 5.9623 (log assets), and mean market-to-book is 2.4504.
- **Investment and financing.** Mean gross investment is 0.1312; mean equity issuance is 0.0268 and mean share repurchase is 0.0178, but both have a median of zero, indicating lumpy external equity transactions.
- **Investor holdings.** The mean portfolio weight is 0.0046, with a highly skewed distribution.
- **Mispricing and dependence variables.** Mean KZ is −0.2549, mean FF is 0.7165, mean ED is −0.0189, and the 95th percentile of price changes is 0.9286.

2.4 Motivating empirical facts before the structural model

- **Equity versus debt financing frequency.** Equity issuance occurs only 14.4% of the time, versus 42.4% for debt issuance. But conditional on issuance, equity raises larger amounts: mean conditional equity issuance is 18.8% of assets versus 12.4% for debt issuance.
- **Repurchases are smoother and more common.** Share repurchases occur 48.6% of the time and average 3.6% of assets conditional on repurchase.
- **Issuance tracks valuation.** In Table 2, equity issuance is positively associated with Tobin's Q, discretionary accruals, and price changes even with firm and year fixed effects. The controlled coefficients are about 0.017 on Q, 0.064 on DACCR, and 0.018 on annual price changes.
- **Investment tracks issuance.** Table 2, Panel B shows that a one percentage point increase in equity issuance is associated with a 15.4 basis point increase in investment in the specification with controls and fixed effects.
- **Heterogeneity by equity dependence.** The issuance-investment link is stronger for firms with higher KZ, lower FF, and higher ED. In other words, equity financing matters most for firms that rely more on equity ex ante.
- **Meme-stock episode as motivation.** In 2021, GameStop, AMC, Bed Bath & Beyond, National Beverage, Koss, and Tesla experienced on average a 73% increase in equity issuance and a 43% increase in investment.
- **Interpretation.** The reduced-form facts are not used as causal evidence. They motivate a structural model where valuation shocks and financing choices interact to shape real investment.

2.5 Additional data for the sources of excess demand

- **Asset-pricing factors.** To study what drives model-implied excess demand, the paper uses 203 cross-sectional asset-pricing factors from Chen and Zimmermann (2022), split into 102 fundamental and 101 non-fundamental factors.
- **Direct sentiment proxies.** The paper also uses RavenPack Composite Sentiment Score (CSS), RavenPack Aggregate Event Volume (AEV), order-imbalance based sentiment (OIB), and earnings-conference-call tone (ECC).
- **Heterogeneity splits.** Firms are sorted by *sentiment risk* (time-series volatility of CSS) and by household ownership share to study which firms face more volatile excess demand.

3 Models and empirical specifications (all core equations + interpretation)

3.1 Production, productivity, and capital accumulation

The firm-level production technology is

$$Y_{n,t} = a_t^{\zeta_n} z_{n,t} K_{n,t}^{\alpha}. \tag{1}$$

Aggregate and idiosyncratic productivity follow AR(1) processes:

$$\log a_{t+1} = \rho_a \log a_t + \nu^a_{t+1}, \qquad \nu^a_{t+1} \sim \mathcal{N}(0, \sigma^2_a), \tag{2}$$

$$\log z_{n,t+1} = \rho_z \log z_{n,t} + \nu^z_{n,t+1}, \qquad \nu^z_{n,t+1} \sim \mathcal{N}(0, \sigma^2_z). \tag{3}$$

Capital evolves according to

$$K_{t+1} = (1-\delta)K_t + I_t, \tag{4}$$

and investment incurs quadratic adjustment costs,

$$\Xi(I_t, K_t) = \xi \frac{I_t^2}{K_t}. \tag{5}$$

- **Interpretation.** The real side is a standard decreasing-returns neoclassical investment model, but the paper later couples it to a demand system for capital providers.
- **Role of ζ_n.** Firms differ in their exposure to aggregate productivity, which later becomes important for the risk-premium channel in misallocation.
- **Role of ξ.** Investment cannot instantly jump to the frictionless optimum; the size of ξ governs how much financing shocks translate into gradual rather than one-shot capital adjustment.

3.2 Cash, debt, and equity financing

Net cash C_t can be positive or negative. Debt is constrained by pledgeable capital:

$$-C_t \le \theta K_t. \tag{6}$$

The firm's funds needed for investment satisfy

$$I_t + \Xi(I_t, K_t) = (1-\tau_c) a_t^{\zeta} z_t K_t^{\alpha} + \delta \tau_c K_t + C_t[1 + r(1-\tau_c)] - \mathring{C}_t - f, \tag{7}$$

and cash after equity-market transactions is

$$C_{t+1} = \mathring{C}_t + E_t - \Psi(E_t) - D_t. \tag{8}$$

When $E_t > 0$, issuance is costly:

$$\Psi(E_t) = (\phi_0 K_t + \phi_1 E_t)\mathbf{1}\{E_t > 0\}. \tag{9}$$

- **Interpretation of C_t.** Positive C_t means net cash; negative C_t means net debt.
- **Interpretation of θ.** This is the collateral parameter. Higher θ means easier borrowing against physical capital.
- **Interpretation of ϕ_0 and ϕ_1.** Equity issuance features both a fixed cost and a marginal cost, capturing underwriting and adverse-selection type frictions.
- **Economic logic.** Even absent investor demand shocks, the firm faces meaningful financing frictions. Investor demand matters because it changes the *effective* cost of external equity on top of these frictions.

3.3 Investor demand for stocks

Investor i's portfolio weight on stock n is

$$w_{i,t}(n) = \frac{\exp\left\{\sum_{k=1}^{K} \beta_{k,i} x_{k,t}(n) + \beta_{0,i}\right\} \epsilon_{i,t}(n)}{1 + \sum_{m \in \mathcal{N}_i \setminus \{0\}} \exp\left\{\sum_{k=1}^{K} \beta_{k,i} x_{k,t}(m) + \beta_{0,i}\right\} \epsilon_{i,t}(m)}. \tag{10}$$

The characteristic vector includes log market equity, log book equity, investment, profitability, dividends, and market beta.

- **Interpretation.** Demand is logit in observed characteristics plus a latent multiplicative shifter $\epsilon_{i,t}(n)$.
- **Latent demand.** $\epsilon_{i,t}(n)$ is the object's central demand shock: investor demand in excess of what fundamentals or characteristics alone would predict.
- **Outside asset.** The outside asset's latent demand is normalized to one.
- **Connection to Koijen-Yogo.** The setup is the firm-side counterpart of the Koijen and Yogo (2019) demand system, but here firm policies are endogenous rather than taken as fixed.

3.4 Market clearing and aggregate demand

Let $p_t(n)$ be log stock price, $s_t(n)$ log shares outstanding, and $A(n)$ the total investable wealth relevant for stock n. Market clearing requires

$$p_t(n) + s_t(n) - \log[A(n) W_t(n)] = 0, \tag{11}$$

where the aggregate portfolio weight on stock n is

$$W_t(n) = \frac{\exp\{\boldsymbol{\beta} \mathbf{x}_t(n)\} \epsilon_t(n)}{1 + \sum_{m \in \mathcal{N}_n \setminus \{0\}} \exp\{\boldsymbol{\beta} \mathbf{x}_t(m)\} \epsilon_t(m)}. \tag{12}$$

The price impact of excess demand and share issuance are defined by

$$\Upsilon^{\epsilon}(\mathbf{x}_t(n), \epsilon_t(n)) \equiv \frac{\partial p_t(n)}{\partial \log \epsilon_t(n)} = -\frac{\partial \log[W_t(n)A(n)]/\partial \log \epsilon_t(n)}{\partial \log[W_t(n)A(n)]/\partial p_t(n) - 1}, \tag{13}$$

$$\Upsilon^{s}(\mathbf{x}_t(n), \epsilon_t(n)) \equiv \frac{\partial p_t(n)}{\partial s_t(n)} = -\frac{\partial \log[W_t(n)A(n)]/\partial s_t(n) - 1}{\partial \log[W_t(n)A(n)]/\partial p_t(n) - 1}. \tag{14}$$

- **Interpretation.** Υ^{ϵ} is the elasticity-like mapping from excess demand into price pressure; Υ^{s} is the price effect of net share issuance.
- **Key novelty.** Because firm policies affect $\mathbf{x}_t(n)$, and $\mathbf{x}_t(n)$ feeds back into investor demand, these price impacts are endogenous objects rather than pure demand-system derivatives.
- **Economic meaning.** Investor demand changes not only valuation but also the terms on which firms can tap or return capital to the equity market.

3.5 Latent demand dynamics and the firm's Bellman problem

Firm-level excess demand evolves as

$$\log \epsilon_{t+1} = \rho_\epsilon \log \epsilon_t + \nu^\epsilon_{t+1}, \qquad \nu^\epsilon_{t+1} \sim \mathcal{N}(0, \sigma^2_\epsilon). \tag{15}$$

The controlling shareholder solves

$$\begin{aligned} V(K, C, z, \epsilon; a, \Gamma) = \max_{\{\Delta s, E, D, I, C'\}} \Big\{ & (1 - \tau_d) D \\ & + \frac{1}{1 + \Delta s} \mathbb{E}_{z,\epsilon,a} \big[m(a, a') V(K', C', z', \epsilon'; a', \Gamma') \big] \Big\}. \end{aligned} \tag{16}$$

The stochastic discount factor is

$$\log m_{t+1} = \log \eta - \gamma_t \nu^a_{t+1} - \frac{1}{2} \gamma_t^2 \sigma_a^2, \qquad \gamma_t = \gamma_0 + \gamma_1 \log a_t. \tag{17}$$

The dollar amount raised by equity issuance satisfies

$$\begin{aligned} E = \Delta s \Bigg\{ 1 + \Bigg[\int_0^{\log \epsilon} \Upsilon^\epsilon(\mathbf{x}_{\Delta s = 0}, \tilde{\epsilon}) \, \mathrm{d} \log \tilde{\epsilon} + \int_0^{\Delta s} \Upsilon^s(\mathbf{x}, \epsilon) \, \mathrm{d} \tilde{s} \Bigg] \Bigg\} \\ \times \Big[V(K, C, z, \epsilon; a, \Gamma) - (1 - \tau_d) D \Big]. \end{aligned} \tag{18}$$

- **Interpretation of the Bellman equation.** Managers maximize the value of the controlling shareholders, not the transacting investors. This is what makes market timing valuable: firms exploit price pressure against outsiders.
- **Interpretation of Δs.** Positive Δs means equity issuance; negative Δs means repurchase.
- **Interpretation of the pricing condition.** The market value of newly issued shares equals the intrinsic value adjusted for the cumulative price pressure induced by investor demand and by the issuance itself.
- **State variables.** The firm's relevant state includes physical capital, cash, idiosyncratic productivity, excess demand, aggregate productivity, and the distribution of firms.

3.6 Recursive competitive equilibrium

A recursive competitive equilibrium consists of

- a value function $V(K, C, z, \epsilon; a, \Gamma)$,
- firm policy functions for K', C', Δs, E, and D,
- investors' portfolio rules $w_{i,t}(n)$,
- and a law of motion for the cross-sectional distribution of firms Γ_t,

such that firms solve the Bellman problem, investors choose portfolios according to the demand system, the asset market clears, and the firm distribution evolves consistently with policy functions and shocks.

- **Why this matters.** The equilibrium has a genuine two-sided feedback loop: investor demand shapes prices and financing; firm policies reshape characteristics and hence future investor demand.

3.7 Auxiliary regression used for indirect inference

The key demand-side identifying equation estimated in the data is

$$\log w_{i,t}(n) \sim \gamma_{me}\, me_{n,t}+\gamma_{be}\, be_{n,t}+\gamma_{prof}\, prof_{n,t}+\gamma_{inv}\, inv_{n,t}+\gamma_{div}\, div_{n,t}+\gamma_{mktbeta}\, mktbeta_{n,t}+\mu_{i,t}(n). \tag{19}$$

- **Interpretation.** If firm characteristics were exogenous, the γ's would directly recover the structural β's. But market equity and other characteristics are endogenous because firms respond to demand shocks.
- **Indirect inference logic.** The paper simulates the model, runs the same auxiliary regression on simulated data, and chooses structural parameters so that simulated auxiliary coefficients match the data.
- **Residual moments.** The autocorrelation and standard deviation of $\mu_{i,t}(n)$ identify ρ_ϵ and σ_ϵ.

3.8 Misallocation accounting

Using the model production function, capital can be written as

$$K_n = \left(\frac{\mathrm{MPK}_n}{\alpha a^{\zeta_n} z_n}\right)^{\frac{1}{\alpha-1}}. \tag{20}$$

Aggregate total factor productivity is

$$\mathrm{TFP} = \frac{Y}{K^\alpha} = \frac{\int_n a^{\zeta_n} z_n \left(\frac{\mathrm{MPK}_n}{\alpha a^{\zeta_n} z_n}\right)^{\frac{\alpha}{\alpha-1}} \mathrm{d}n}{\left[\int_n \left(\frac{\mathrm{MPK}_n}{\alpha a^{\zeta_n} z_n}\right)^{\frac{1}{\alpha-1}} \mathrm{d}n\right]^\alpha}. \tag{21}$$

The welfare-style misallocation statistic is

$$\mathrm{TFP}_{loss} = \log(\mathrm{TFP}_e) - \log(\mathrm{TFP}). \tag{22}$$

- **Interpretation.** In an efficient allocation, MPK is equalized across firms. Any extra dispersion in log(MPK) is interpreted as capital misallocation.
- **Why use both** $\mathrm{Var}(\log \mathrm{MPK})$ **and** TFP_{loss}**.** If the joint distribution is not exactly log-normal, variance of $\log \mathrm{MPK}$ is not a sufficient statistic; the TFP-loss measure is the more general object.

3.9 Robustness extensions in the model

The paper also considers non-convex capital adjustment costs,

$$\Xi(I_t, K_t) = \left(\Upsilon^+ K_t + \xi^+ \frac{I_t^2}{K_t}\right) \mathbf{1}\{I_t \geq 0\} + \left(\Upsilon^- K_t + \xi^- \frac{I_t^2}{K_t}\right) \mathbf{1}\{I_t < 0\}, \tag{23}$$

and correlated productivity and demand shocks,

$$\begin{bmatrix} \log z_{t+1} \\ \log \epsilon_{t+1} \end{bmatrix} = \begin{bmatrix} \rho_z & \rho_{z\epsilon} \\ \rho_{\epsilon z} & \rho_\epsilon \end{bmatrix} \begin{bmatrix} \log z_t \\ \log \epsilon_t \end{bmatrix} + \begin{bmatrix} \sigma_z & 0 \\ 0 & \sigma_\epsilon \end{bmatrix} \varepsilon_{t+1}. \tag{24}$$

- **Interpretation.** These extensions test whether the baseline misallocation result depends on especially simple adjustment-cost assumptions or on the assumption that excess demand is independent of future fundamentals.
- **Bottom line.** The paper reports that the main quantitative conclusions survive these modifications.

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline identification idea

- **Two-sided equilibrium variation.** The paper identifies investor demand and firm policy jointly. Investors choose portfolios as functions of firm characteristics; firms know they face time-varying demand and choose policies accordingly.
- **Core challenge.** Market equity, investment, dividends, and other characteristics are endogenous because they are themselves policy outcomes. Therefore, directly regressing holdings on characteristics would confound demand with supply-side responses.
- **Solution.** Use indirect inference: match the *biased* auxiliary relationship in the data with the same relationship generated by the model, so the endogenous feedback loop is reproduced rather than ignored.

4.2 Demand-side identification

- **Structural demand parameters.** $\boldsymbol{\beta} = \{\beta_{me}, \beta_{be}, \beta_{prof}, \beta_{inv}, \beta_{div}, \beta_{mktbeta}\}$ govern how holdings respond to log market equity, log book equity, profitability, investment, dividends, and market beta.
- **Moments used.** The six coefficients from the auxiliary regression identify the six demand loadings, while the residual autocorrelation and residual standard deviation identify ρ_ϵ and σ_ϵ.
- **Why not use a plain IV baseline?** Koijen and Yogo (2019) use an IV strategy based on investor investment universes. This paper instead chooses indirect inference because firm policies respond to latent demand and those responses alter the characteristics that enter the demand system.

4.3 Firm-side SMM identification

- **Idiosyncratic productivity process.** The autocorrelation and standard deviation of firms' income-to-assets identify ρ_z and σ_z.
- **Capital accumulation.** Mean investment identifies the depreciation rate δ, while the standard deviation of investment identifies the capital adjustment cost ξ.
- **Equity issuance costs.** Mean and standard deviation of equity issuance identify ϕ_1, and issuance frequency identifies the fixed issuance cost ϕ_0.
- **Repurchases and payouts.** The mean and standard deviation of repurchases plus mean dividend yield help pin down payout behavior.
- **Borrowing friction.** Average leverage identifies the collateral parameter θ.
- **Technology and fixed costs.** Mean income-to-assets and mean market-to-book help pin down α and f.

4.4 Calibrated aggregate objects

- **Taxes and interest-rate environment.** The paper sets the corporate tax rate $\tau_c = 0.35$, dividend tax rate $\tau_d = 0.15$, and $\eta = 0.98$ to match an average risk-free rate near 2%.
- **Investor granularity.** The number of holdings per investor is set to $J = 1000$, matching the observed average portfolio weight.
- **Aggregate productivity.** $\rho_a = 0.8145$ and $\sigma_a = 0.0140$ are calibrated from standard business-cycle values.
- **Risk-price parameters.** γ_0, γ_1, and σ_ζ are calibrated to match the average equity premium (7.69%), average Sharpe ratio (0.174), and the cross-sectional dispersion of expected stock returns (0.221).

4.5 Validation and external discipline

- **Untargeted policy functions.** The model reproduces how net issuance and investment covary with income-to-assets, market-to-book, and leverage. Removing excess demand worsens the fit sharply.
- **Excess demand and issuance.** Model-implied excess demand predicts actual equity issuance strongly: an interquartile increase in ϵ is associated with a 1.46% increase in issuance in the model versus 1.12% in the data.
- **SEO overvaluation check.** The model implies average mispricing of 33.85% for equity issuers, close to the roughly 30% overvaluation benchmark from the SEO literature.
- **Alternative estimation methods.** Re-estimating demand with Koijen-Yogo style IV or van der Beck style flow moments leaves the qualitative and quantitative misallocation results broadly intact.

5 Tables and figures

5.1 Table 1: Variable definitions

Read:

- This is the variable dictionary for both the reduced-form evidence and the structural estimation.
- The important objects are the financing variables (equity issuance, debt issuance, repurchase), valuation variables (market equity, market-to-book, dividends, price changes), and dependence measures (KZ, FF, ED).
- The table shows clearly that the investor-demand system is disciplined with *observable* firm characteristics rather than arbitrary latent states alone.

Table 1
Variable definitions.

Variable	Construction
Asset growth	Change in total assets from the previous year divided by total assets in the previous year
Book equity	Book value of stockholders' equity, plus deferred taxes and investment tax credits (if available), minus the book value of preferred stock
DACCR	Discretionary accruals as in Polk and Sapienza (2008) and Chan et al. (2001)
Debt issuance	Change in total debt divided by total assets (if greater than 2%)
Dividend yield	Dividends on common and preferred stock divided by market equity
ED	The difference between the average equity issuance-to-asset ratio and the debt issuance-to-asset ratio over the past five years
Equity issuance	Sale of common and preferred stock (if greater than 2% following the algorithm in McKeon (2015)) divided by total assets
FF	Financial flexibility index as in Doidge et al. (2009)
Firm size	Log of total assets
Gross investment	Capital expenditures minus sales of property, divided by gross plant, property and equipment
Income-to-assets ratio	Operating income before depreciation and amortization divided by total assets
Investment universe	Stocks held in the current quarter and the previous 12 quarters, following Koijen and Yogo (2019)
KZ	The four-variable version of KZ-index (Kaplan and Zingales, 1997) following Baker et al. (2003)
Leverage	Total debt divided by total assets
Market-to-book ratio	Total assets minus book equity plus market equity, divided by total assets
Market equity	Price times shares outstanding
Operating profitability	Revenues minus the cost of goods sold, interest expenses, and selling, general, and administrative expenses divided by book equity
Outside asset	Stocks with missing characteristics or returns or stocks with CRSP share codes other than 10 or 11, following Koijen and Yogo (2019)
Portfolio weight	The ratio of the dollar amount invested in a given stock to the total dollar amount invested in all stocks held by an institution
Price changes	Changes in stock prices divided by stock price in the previous year
Share repurchase	Purchase of common and preferred stock divided by total assets

Table 2
Motivating empirical facts.

Panel A: Equity issuance	(1)	(2)	(3)	(4)	(5)	(6)
Q	0.019***	0.017***				
	(0.001)	(0.002)				
DACCR			0.060***	0.064***		
			(0.009)	(0.010)		
Price changes					0.018***	0.018***
					(0.002)	(0.002)
Other firm controls	NO	YES	NO	YES	NO	YES
Year FE	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES
Adjusted R^2	0.391	0.403	0.370	0.384	0.377	0.390
Panel B: Investment	(1)	(2)	(3)	(4)	(5)	(6)
Equity issuance	0.204***	0.154***	0.274***	0.170***	0.293***	0.125***
	(0.016)	(0.018)	(0.017)	(0.020)	(0.028)	(0.018)
Equity issuance × KZ				0.015**		
				(0.008)		
Equity issuance × FF					−0.116***	
					(0.016)	
Equity issuance × ED						0.239**
						(0.109)
Other firm controls	NO	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	NO	YES	YES	YES
Adjusted R^2	0.322	0.374	0.166	0.375	0.397	0.367

This table reports the relationship between firms' investment, equity issuance, and mispricing measures including Tobin's Q, discretionary accruals (DACCR), and price changes. In Panel A, the dependent variable is the equity issuance to asset ratio. In Panel B, the dependent variable is the firm investment rate. The sample period is 1980–2021. We exclude firms in the utility (SIC 4900-4999) and financial (SIC 6000-6999) sectors. Other firm controls include firm size, firm age, cash flow to asset ratio, and debt issuance. KZ is the four-variable version of the KZ-index (Kaplan and Zingales, 1997), following the methodology of Baker et al. (2003). FF represents the Financial Flexibility index as defined in Doidge et al. (2009). ED is the difference between the average equity issuance-to-asset ratio and the debt issuance-to-asset ratio over the past five years. Standard errors are clustered at the firm level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

5.2 Table 2: Motivating empirical facts

Read:

- **Panel A.** Firms issue more equity when they look more overvalued: high Q, high discretionary accruals, and high stock-price appreciation all predict issuance.
- **Panel B.** Investment rises with equity issuance, and the slope is steeper for firms more dependent on equity finance.
- **Takeaway.** These are the key reduced-form facts the model is built to rationalize: equity-market conditions move real investment, and the response is heterogeneous across firms.

5.3 Table 3: Summary statistics

Read:

- Most firms do not issue equity or repurchase in a typical year, but the right tail is substantial, so a lumpy policy model is appropriate.
- Mean market-to-book of 2.45 and wide dispersion in price changes suggest valuation shocks are economically important.
- The financing-dependence proxies and holdings weights exhibit meaningful cross-sectional variation, which is important for identification.

Table 3
Summary statistics.

Variable	Mean	P5	P25	P50	P75	P95
Income-to-assets ratio	0.1367	−0.0611	0.0858	0.1428	0.2053	0.3343
Leverage	0.2367	0.0000	0.0902	0.2239	0.3521	0.5611
Firm size	5.9623	2.5859	4.3891	5.8632	7.4630	9.6650
Log book equity	5.2192	1.8776	3.7365	5.1676	6.6785	8.7439
Log market equity	5.7482	2.2739	4.0313	5.6276	7.3136	9.6885
Pct. of equity issuance	0.0268	0.0000	0.0000	0.0000	0.0000	0.1812
Pct. of share repurchase	0.0178	0.0000	0.0000	0.0000	0.0106	0.0803
Gross investment	0.1312	0.0096	0.0536	0.0926	0.1555	0.3786
Asset growth	0.1199	−0.1793	−0.0152	0.0608	0.1637	0.6052
Market-to-book ratio	2.4504	0.6633	1.0430	1.5085	2.6510	8.0230
Dividend yield	0.0231	0.0000	0.0004	0.0138	0.0296	0.0680
Portfolio weight	0.0046	0.0000	0.0001	0.0012	0.0041	0.0151
KZ	−0.2549	−2.5001	−0.8103	−0.0962	0.5625	1.4222
FF	0.7165	0.0000	0.0000	1.0000	1.0000	2.0000
ED	−0.0189	−0.1235	−0.0469	−0.0155	0.0000	0.0782
Debt issuance	0.0423	0.0000	0.0000	0.0000	0.0467	0.2172
DACCR	0.0001	−0.1513	−0.0370	0.0034	0.0406	0.1409
Price changes	0.0670	−0.5194	−0.1932	0.0336	0.2822	0.9286

This table reports summary statistics for our sample. The sample period is 1980–2021. We exclude firms in the utility (SIC 4900–4999) and financial (SIC 6000–6999) sectors. The balance-sheet variables are obtained from Compustat, stock prices are obtained from CSRP, and portfolio weights are calculated using the Form 13F data. All variables are defined in Table 1.

Table 4
Parameter estimates.

Panel A: Calibrated parameters		Value	
τ_c	Corporate tax rate	0.35	
τ_d	Dividend tax rate	0.15	
J	Number of holdings	1000	
η	SDF parameter	0.98	
γ_0	SDF parameter	0.2264	
γ_1	SDF parameter	−12.5511	
ρ_a	Persistence of aggregate productivity shocks	0.8145	
σ_a	Standard deviation of aggregate productivity shocks	0.0140	
σ_ζ	Standard deviation of exposure to aggregate productivity shocks	0.2187	
Panel B: Parameters governing investors' demand		**Value**	**Std. Err.**
β_{me}	Demand coefficient on log market equity	0.3416	0.0060
β_{be}	Demand coefficient on log book equity	0.1134	0.0029
β_{prof}	Demand coefficient on profitability	0.1780	0.0213
β_{inv}	Demand coefficient on investment rate	0.0194	0.0056
β_{div}	Demand coefficient on dividend	0.3255	0.0097
$\beta_{mktbeta}$	Demand coefficient on market beta	−0.0348	0.0050
ρ_ϵ	Auto-correlation of excess demand	0.8068	0.0092
σ_ϵ	Standard deviation of excess demand	0.0931	0.0011
Panel C. Parameters governing firms' operations		**Value**	**Std. Err.**
α	Curvature of production function	0.6203	0.0038
ρ_z	Persistence of idiosyncratic productivity shocks	0.8307	0.0218
σ_z	Standard deviation of idiosyncratic productivity shocks	0.0904	0.0064
δ	Depreciation rate	0.1509	0.0005
ξ	Capital adjustment cost	0.3774	0.0197
θ	Collateral constraint	0.8299	0.0178
ϕ_1	Linear equity issuance cost	0.0968	0.0069
ϕ_0	Fixed equity issuance cost	0.0109	0.0008
f	Fixed operating cost	1.7484	0.0407

This table reports the model parameter estimates. Panel A presents calibrated parameters, Panel B presents parameter estimates for investor demand, and Panel C presents parameter estimates for firms' operations. Standard errors for the estimated parameters are clustered at the firm level and reported in the last column.

5.4 Table 4: Parameter estimates

Read:

- **Demand side.** The most important coefficient is $\beta_{me} = 0.3416$, implying fairly inelastic investor demand. $\rho_\epsilon = 0.8068$ and $\sigma_\epsilon = 0.0931$ imply persistent and economically large excess-demand shocks.
- **Firm side.** The technology curvature is $\alpha = 0.6203$, depreciation is $\delta = 0.1509$, adjustment costs are $\xi = 0.3774$, collateralability is $\theta = 0.8299$, and issuance costs are $\phi_1 = 0.0968$ and $\phi_0 = 0.0109$.
- **Interpretation.** The estimates are in a realistic range and, crucially, support a world where demand shocks can create sizable price pressure without absurd volatility.

5.5 Table 5: Moment conditions

Read:

- The model is estimated in an over-identified system, so fit is nontrivial.
- Simulated moments are reasonably close to actual moments for both the auxiliary demand coefficients and the firm-level balance-sheet moments.
- The table is the main evidence that the estimated model replicates the joint behavior of holdings, issuance, repurchases, leverage, investment, profitability, and valuation.

5.6 Table 6: Untargeted moments

Read:

- The baseline model reproduces the qualitative dependence of net issuance and investment on transformed state variables such as income-to-assets, beginning market-to-book, and leverage.

Table 5
Moment conditions.

	Actual	Simulated	Std. Err.
(1) γ_{me} in the auxiliary equation	0.8256	0.7458	0.0175
(2) γ_{be} in the auxiliary equation	0.0207	0.0255	0.0013
(3) γ_{prof} in the auxiliary equation	0.1161	0.1142	0.0025
(4) γ_{inv} in the auxiliary equation	0.0996	0.1049	0.0028
(5) γ_{div} in the auxiliary equation	0.3669	0.4069	0.0276
(6) $\gamma_{mktbeta}$ in the auxiliary equation	−0.0541	−0.0761	0.0012
(7) Auto-correlation of residual in the auxiliary equation	0.7555	0.8932	0.0361
(8) Std. dev. of residual in the auxiliary equation	0.0925	0.0769	0.0014
(9) Auto-correlation of income-to-assets ratio	0.7961	0.8550	0.0155
(10) Std. dev. of income-to-assets ratios	0.0894	0.0648	0.0066
(11) Percentage of equity issuance	0.0268	0.0310	0.0013
(12) Std. dev. of equity issuance	0.0659	0.0787	0.0063
(13) Frequency of equity issuance	0.1443	0.1726	0.0038
(14) Percentage of share repurchase	0.0178	0.0141	0.0005
(15) Std. dev. of share repurchase	0.0494	0.0414	0.0044
(16) Mean investment	0.1312	0.1554	0.0019
(17) Std. dev. of investment	0.1263	0.0967	0.0056
(18) Mean leverage	0.2367	0.2461	0.0023
(19) Mean income-to-assets ratio	0.1367	0.1577	0.0021
(20) Mean market-to-book ratio	2.4504	3.1366	0.0242
(21) Mean dividend yield	0.0231	0.0326	0.0004

This table reports the empirical (*Actual*) and simulated (*Simulated*) moments that we use in our SMM estimation. Moments (1)–(6) are the regression coefficients of the auxiliary model in Eq. (21). Moments (7) and (8) are the autocorrelation and standard deviation of residuals of Eq. (21). Moments (9)–(21) correspond to autocorrelation, averages, and standard deviations of firms' balance-sheet variables. In the last column (*Std. Err.*), we report standard errors of the moment conditions.

Table 6
Untargeted moments.

	Data (1)	Baseline model (2)	Counterfactual (3)
Net issuance on			
Income-to-assets ratio	−0.1222	−0.1366	−0.0646
Beginning-of period market-to-book ratio	0.1031	0.0564	−0.2264
Beginning-of period leverage	0.0970	0.0802	0.0352
Investment on			
Income-to-assets ratio	0.0590	0.0692	0.2712
Beginning-of period market-to-book ratio	0.1912	0.0446	−0.0848
Beginning-of period leverage	−0.1090	−0.1022	0.0810

This table presents the fit of the model's empirical policy functions, which describe the relationship between the (transformed) firm state variables and the optimal policies in that state following Bazdresch et al. (2018). The observable state variables include firm income-to-assets ratio, market-to-book ratio, and leverage. We focus on firms' investment and equity issuance policies. Following Bazdresch et al. (2018), we first transform the raw variables into quartiles based on their ranking within a given year and then estimate the regression coefficients using these quartile variables as inputs. Column (1) presents the results calculated using the actual data, column (2) contains that from the simulation of the baseline model, and column (3) contains the results from a counterfactual model where we set the mean and standard deviation of $\log(\epsilon)$ to zero.

- The counterfactual model without excess demand fails badly, often flipping the sign of key policy gradients.
- This table is an important external-validity check: investor demand is needed not just to fit targeted moments but to replicate how firms behave across states.

5.7 Figure 1: Excess demand and price impact

Read:

- The figure is upward sloping: stronger excess demand raises the premium of market price over intrinsic value; weaker demand creates discounts.
- Around zero excess demand, price impact is near zero; by $\log \epsilon \approx 0.3$, the premium is large and positive.
- This is the simplest visual summary of the market-timing mechanism: equity demand directly changes the valuation terms on which firms transact.

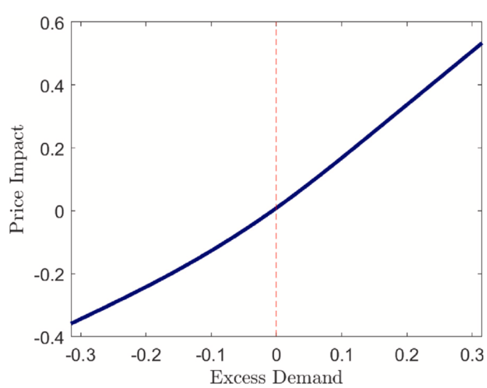

Fig. 1. Excess demand and price impact.
This figure presents the model-predicted relationship between excess demand and the price impact on stocks. The solution is calculated using the parameter estimates reported in Table 4. Price impact is defined as the percentage difference between a firm's market price and its intrinsic value ex-dividend, $V-(1-\tau_d)D$, with V as defined in Eq. (16). We calculate the average price impact across all state variables (except ϵ) using the ergodic distribution.

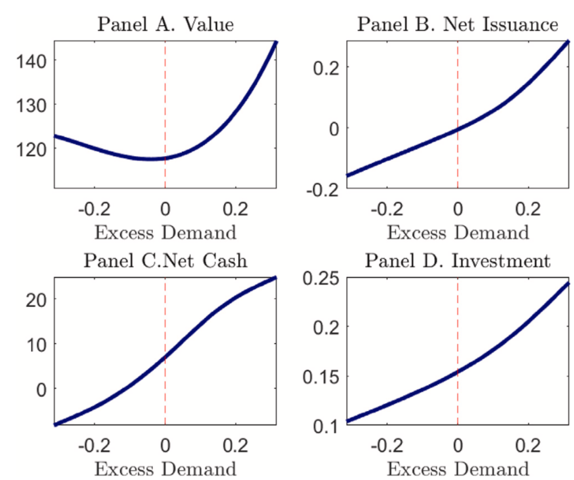

Fig. 2. Excess demand, firm value, and optimal policies.
This figure illustrates the model-predicted relationships among excess demand, a firm's intrinsic value as defined in Eq. (16), and its optimal investment and financing policies. The solution is calculated using the parameter estimates reported in Table 4. For the firm value and policies, we calculate the average across all state variables (except ϵ) using the ergodic distribution.

Table 7
Comparison of alternative estimation strategies.

	OLS Reg (1)	Baseline (2)	IV-holding based (3)	IV-flow based (4)
Relationship btw demand and:				
Market equity	0.8256	0.3416	0.6508	0.3500
Book equity	0.0207	0.1134	0.1572	0.3058
Profit	0.1161	0.1780	0.1331	0.1279
Investment	0.0996	0.0194	0.2487	0.4676
Dividends	0.3669	0.3255	2.0878	4.4616
Market beta	−0.0541	−0.0348	−0.0127	−0.0093
Auto-correlation of residual	0.7555	0.8068	0.6617	0.7773
Stdev. of residual	0.0925	0.0931	0.1472	0.1746

This table reports demand-side parameter estimates using three different approaches. In column (1), we report the regression coefficient, γ, by estimating the auxiliary Equation (21) via OLS; in column (2), we report the estimated structural parameters, β in Eq. (10), using the indirect inference method described in Section 4.2; in column (3), we report the coefficient estimates using the instrumental variable approach of Koijen and Yogo (2019) based on cross-sectional regression of investors' holdings; in column (4) we report the parameter estimates using the flow-based approach of van der Beck (2022).

5.8 Figure 2: Excess demand, firm value, and optimal policies

Read:

- **Panel A (Value).** Intrinsic value is U-shaped in excess demand: very negative demand is good because the firm can repurchase cheaply; very positive demand is good because the firm can issue dearly.
- **Panel B (Net issuance).** Net issuance increases monotonically with excess demand and crosses zero near $\log \epsilon = 0$.
- **Panel C (Net cash).** Cash holdings rise with excess demand because positive demand lets firms issue and accumulate liquidity.
- **Panel D (Investment).** Investment rises with excess demand because positive demand relaxes financing conditions, while negative demand forces repurchases and crowds out capital expenditure.

5.9 Table 7: Comparison of alternative estimation strategies

Read:

- Plain OLS on the auxiliary regression yields a much larger market-equity loading (0.8256) than the structural estimate (0.3416).
- Koijen-Yogo style IV and van der Beck style flow-based estimates correct the bias in the same direction: the market-equity coefficient is far below the OLS value once endogeneity is addressed.
- The message is methodological: demand elasticities that ignore endogenous prices and firm responses look too flat.

5.10 Table 8: Investor demand, financial constraints, and the flow of funds

Read:

- Low-demand firms repurchase shares; high-demand firms issue equity. This pattern holds for both constrained and less constrained firms.
- Among constrained firms, investment's share of funds rises from 32.71% in low-ϵ states to 39.68% in high-ϵ states.
- Among less constrained firms, the extra proceeds from favorable demand are used more for dividends and cash retention than for additional investment.

Table 8
Investor demand, financial constraints, and the flow of funds.

	Constrained		Less constrained	
	Low ϵ (1)	High ϵ (2)	Low ϵ (3)	High ϵ (4)
Sources				
Profit	0.8252	0.7386	0.7187	0.6398
Net cash reduction	0.1512	0.0907	0.2069	0.0507
Interest income	0.0001	0.0309	0.0449	0.0677
Equity issuance	0.0235	0.1397	0.0294	0.2418
Uses				
Investment	0.3271	0.3968	0.3781	0.3819
Adjustment cost	0.0240	0.0277	0.0247	0.0245
Taxes	0.1373	0.1299	0.1315	0.1179
Cash increase	0.1077	0.1120	0.0529	0.0934
Interest payment	0.0386	0.0123	0.0040	0.0002
Dividends	0.0632	0.1500	0.0955	0.2422
Repurchases	0.0816	0.0000	0.1373	0.0000
Fixed operating costs	0.2204	0.1714	0.1760	0.1401

This table presents the decomposition of the sources and uses of funds for simulated firms based on our model. We calculate the fraction of each source (top four rows) or use (bottom eight rows) relative to the total and report the average values within each subsample of firms. Firms are first divided into "High" or "Low" ϵ based on whether their current period $log(\epsilon)$ is above or below zero. Within each group, firms are further split based on whether their distance to the collateral constraint ($\theta - \frac{-C}{K}$) is above or below the median. We classify firms as "Constrained" if their distance to the collateral constraint is below the median and "Less Constrained" if otherwise. Model simulations are based on parameter values reported in Table 4.

Table 9
Investor clientele and firm policies.

Δ Investors	Mkt cap (1)	Net Issuance (2)	Investment (3)	MPK (4)
X = 0% (Baseline)	1.0000	0.0169	0.1509	0.1783
X=10%	1.0921	0.0427	0.1545	0.1767
20%	1.1842	0.0685	0.1582	0.1751
30%	1.2766	0.0944	0.1618	0.1734
40%	1.3687	0.1202	0.1654	0.1718
50%	1.4613	0.1461	0.1691	0.1701
60%	1.5535	0.1719	0.1727	0.1685
70%	1.6457	0.1977	0.1764	0.1669
80%	1.7380	0.2236	0.1800	0.1652
90%	1.8304	0.2494	0.1837	0.1636

This table presents how market valuation, equity issuance, and return on investment would change for firms in our model as their investor clientele changes. In the first row, we present the results of the baseline economy without changes in investor clientele. For each of the remaining rows, we randomly select a subsample of firms and swap X percent of their investors with "meme stock" investors, whose ϵ matches that of GameStop, AMC, Bed, Bath & Beyond, National Beverage, Koss, and Tesla during 2021. We report firms' average market capitalization (scaled by that in the baseline model) in column (1), the average percentage of net equity issuance (equity issuance minus share repurchases) in reported column (2), and firms' average investment (scaled by the level of capital stock in the baseline model) is in column (3), and the return of investment (MPK) is in column (4).

- So the real crowding-out channel is strongest when investor demand interacts with tight financing constraints.

5.11 Table 9: Investor clientele and firm policies

Read:

- Replacing larger fractions of a firm's investors with "meme-stock" investors sharply raises market capitalization, net issuance, and investment.
- With 90% meme-stock investors, market capitalization rises to 1.8304 times baseline, net issuance to 0.2494, and investment to 0.1837.
- But MPK falls from 0.1783 to 0.1636, showing that sentiment-driven financing induces low-return investment as well as more investment.

5.12 Table 10: Investor demand and capital misallocation

Read:

- In the baseline parameterization, removing excess demand lowers Var(log MPK) from 0.0296 to 0.0217, a 26.9% decline, and lowers TFP loss from 2.28% to 1.75%, a 23.36% decline.
- When investor demand becomes more passive-like (β_{me} closer to one), the price-impact channel becomes even stronger conditional on a given amount of excess demand.
- The table illustrates the core aggregate punchline: dispersion in investor demand is a quantitatively important source of capital misallocation.

5.13 Table 11: Comparing sources of capital misallocation

Read:

- Shutting down excess demand produces a much larger improvement in allocation than relaxing debt-market frictions or eliminating convex adjustment costs.
- Specifically, reduced debt frictions lower Var(log MPK) only 6.91%, and removing adjustment costs lowers it only 3.04%, versus 26.9% when excess demand is removed.
- The risk-premium channel is more important than debt or adjustment frictions, but the excess-demand channel is still at least comparable and in this paper larger.

Table 10
Investor demand and capital misallocation.

Panel A: baseline β				
	$\log(\epsilon)=$ baseline	$\log(\epsilon)=50\%\times$ baseline	$\log(\epsilon)=0$	Percent change (from baseline to 0)
(1) Var(log(MPK))	0.0296	0.0258	0.0217	−26.90%
(2) TFP loss	2.28%	2.06%	1.75%	−23.36%
Panel B: 50% passive investor β				
	$\log(\epsilon)=$ baseline	$\log(\epsilon)=50\%\times$ baseline	$\log(\epsilon)=0$	Percent change (from baseline to 0)
(1) Var(log(MPK))	0.0362	0.0297	0.0217	−40.22%
(2) TFP loss	2.75%	2.30%	1.75%	−36.42%
Panel C: 90% passive investor β				
	$\log(\epsilon)=$ baseline	$\log(\epsilon)=50\%\times$ baseline	$\log(\epsilon)=0$	Percent change (from baseline to 0)
(1) Var(log(MPK))	0.0409	0.0346	0.0217	−47.04%
(2) TFP loss	2.98%	2.63%	1.75%	−41.30%

This table presents the variance of the log(MPK) and the TFP loss for firms simulated under various scenarios based on different values of demand parameters (β) and excess demand shocks. In Panel A, investors' demand parameters β are maintained at the baseline estimates reported in Table 4. We consider counterfactual scenarios where the mean (in logs) and standard deviation of excess demand shocks are set to either 50% of the baseline estimate or 0. The last column calculates the percentage differences in the variables of interest. We repeat the exercise in Panel B, where we set β to the average between the baseline estimates and that of passive investors, and in Panel C where we set β as a convex combination of 90% of passive investors' β and 10% of the baseline estimates. Passive investors have β equal to 1 for market equity and 0 for all other characteristics.

Table 11
Comparing sources of capital misallocation.

Panel A: Effect of latent demand			
	Baseline	No excess demand	Percentage change
(1) Var(log(MPK))	0.0296	0.0217	−26.90%
(2) TFP loss	2.28%	1.75%	−23.36%
Panel B: Effect of debt market frictions			
		Reduced debt frictions	Percentage change
(1) Var(log(MPK))		0.0276	−6.91%
(2) TFP loss		2.22%	−2.74%
Panel C: Effect of capital adjustment cost			
		No adjustment costs	Percentage change
(1) Var(log(MPK))		0.0287	−3.04%
(2) TFP loss		2.17%	−5.14%
Panel D: Effect of risk premium			
		No risk premium	Percentage change
(1) Var(log(MPK))		0.0253	−14.56%
(2) TFP loss		2.02%	−11.35%

This table presents the variance of the log(MPK) and the TFP loss for firms under the baseline model and several counterfactual scenarios. Panel A reports the results from the baseline model and the counterfactual scenario where we switch off excess demand shocks by setting the mean (in logs) and standard deviation of excess demand shocks to 0 while keeping the value of other parameters the same as the baseline case. Panel B reports the results from the counterfactual exercise where we reduce debt market frictions by setting the collateral constraint parameter θ to 3. Panel C reports the results from the counterfactual exercise where we set the capital adjustment cost parameter ξ to 0. Panel D reports the results from the counterfactual exercise where we remove firms' exposure to the aggregate risk. Column "Percentage change" reports the percentage changes in our variables of interest from the baseline model to the respective counterfactual cases.

Table 12
Subsample estimation.

	Low SR (1)		High SR (2)		Low HH (3)		High HH (4)	
Panel A: Parameters governing investors' demand								
	Value	Std. Err.	Value	Std. Err.	Value	Std. Err.	Value	Std. Err.
β_{me}	0.3466	0.0076	0.3142	0.0122	0.3453	0.0193	0.3193	0.0280
β_{be}	0.1158	0.0018	0.1152	0.0041	0.1132	0.0048	0.1135	0.0026
β_{prof}	0.1763	0.0107	0.1750	0.0191	0.1786	0.0086	0.1796	0.0463
β_{inv}	0.0195	0.0022	0.0194	0.0022	0.0195	0.0026	0.0193	0.0078
β_{div}	0.3284	0.0106	0.3262	0.0390	0.3271	0.0136	0.3231	0.0293
$\beta_{mktbeta}$	−0.0348	0.0209	−0.0350	0.0096	−0.0349	0.0206	−0.0345	0.0240
ρ_{ϵ}	0.8102	0.0059	0.8138	0.0060	0.8147	0.0141	0.8154	0.0069
σ_{ϵ}	0.0908	0.0014	0.1028	0.0012	0.0920	0.0059	0.1017	0.0045
Panel B. Parameters governing firms' operation								
α	0.6223	0.0041	0.6191	0.0047	0.6221	0.0024	0.6223	0.0084
ρ_z	0.8262	0.0132	0.8338	0.0664	0.8276	0.0257	0.8193	0.0364
σ_z	0.0904	0.0024	0.0912	0.0040	0.0900	0.0030	0.0913	0.0035
δ	0.1527	0.0005	0.1531	0.0010	0.1514	0.0005	0.1521	0.0012
ξ	0.3800	0.0209	0.3807	0.0607	0.3769	0.0110	0.3784	0.0429
θ	0.8340	0.0146	0.8261	0.0535	0.8311	0.0109	0.8155	0.0344
ϕ_1	0.0964	0.0024	0.0964	0.0062	0.0972	0.0079	0.0967	0.0073
ϕ_0	0.0109	0.0007	0.0109	0.0018	0.0109	0.0011	0.0108	0.0019
f	1.7461	0.0142	1.6815	0.0502	1.7029	0.0275	1.7735	0.0619
Panel C. Price impacts								
$\frac{\partial p}{\partial \log \epsilon}$	1.6199		1.5206		1.6171		1.5381	
$\frac{\partial p}{\partial s}$	−0.3184		−0.3322		−0.3290		−0.3447	
Panel D. Effect of shutting down excess demand								
Δ%Var(log(MPK))	−25.60%		−31.88%		−23.98%		−29.23%	
Δ%TFP loss	−21.73%		−27.64%		−19.77%		−24.99%	

This table presents the model parameter estimates, price impacts, and percentage changes in capital misallocation measures when we shut down investors' excess demand in different subsamples of firms. "High SR" ("Low SR") is the subsample of firms with above (below) median sentiment risk, and "High HH" ("Low HH") is the subsample of firms with above (below) median percentages of household holdings. Panels A and B present results for parameters that dictate investor demand and firms' operations, respectively. The definitions of these parameters are provided in Table 4. Panel C reports the average elasticities of price to excess demand and equity issuance as defined in Eqs. (13) and (14). Panel D reports the percentage changes in the variance of log(MPK) and the TFP loss when we remove excess demand by setting the mean (in logs) and standard deviation of excess demand shocks to zero while keeping all other parameters at the estimated levels specific to each subsample.

5.14 Table 12: Subsample estimation

Read:

- High-sentiment-risk firms and high-household-ownership firms face more volatile latent demand: $\sigma_\epsilon = 0.1028$ for High SR versus 0.0908 for Low SR, and 0.1017 for High HH versus 0.0920 for Low HH.
- When excess demand is shut down, the reduction in misallocation is larger for these subsamples: for High SR firms, Var(log MPK) falls 31.88% and TFP loss falls 27.64%.
- This is strong evidence that sentiment-sensitive investors are the economically important carriers of the demand shocks emphasized by the paper.

6 Short summary

- **Conclusion.** Investor demand materially affects firms' financing terms, investment decisions, and the aggregate allocation of capital.
- **Intuition.** Positive demand subsidizes equity issuance and relaxes financing constraints, while negative demand encourages repurchases and crowds out investment. Because these effects differ across firms, the cross-sectional dispersion in MPK rises.
- **Empirical lesson.** To quantify the real effects of equity-market mispricing or sentiment, one must model the joint equilibrium between investor demand and firm policy rather than treating firm characteristics or share supply as fixed.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Demand shocks in equity markets are not innocuous valuation phenomena; they change firms' real financing and investment behavior.
- **Magnitude.** Excess demand accounts for roughly one-quarter of observed dispersion in log(MPK) and one-quarter of TFP losses in the model.
- **Credibility.** The demand system is disciplined with holdings data, the firm side is disciplined with balance-sheet moments, and the model passes untargeted policy-function and SEO-overvaluation checks.
- **Mechanism evidence.** Positive demand causes issuance, cash accumulation, and more investment; negative demand causes repurchases and lower investment, especially among constrained firms.
- **Sentiment channel.** Model-implied excess demand lines up more with non-fundamental and sentiment variables than with fundamental accounting signals, and its real effects are stronger where sentiment-sensitive investors matter more.
- **Robustness.** The main quantitative message survives alternative demand-estimation strategies and alternative financing or adjustment-cost specifications.

7.2 Economic intuition

- Investor demand acts like a time-varying wedge in the effective cost of equity capital.
- Firms do not passively receive those wedges: they actively time the market, choosing issuance or repurchase in response to price pressure.
- Because some firms receive cheap capital while others are induced to divert resources toward repurchases, demand shocks widen the cross-sectional dispersion in marginal investment returns.
- The same demand force can be individually helpful and socially distortive: it relaxes some constraints on average, yet it still worsens aggregate allocation because it pushes capital toward firms with favorable investor sentiment rather than toward firms with the highest MPK.

7.3 Takeaway

This paper's big payoff is to show that investor demand belongs inside the resource-allocation conversation, not only inside asset pricing. Once firms can issue or repurchase equity in response to demand, sentiment-driven valuation wedges become real financing wedges. The paper's structural contribution is to quantify that channel and show that it is large enough to compete with more standard sources of capital misallocation.